```
import numpy as np
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense
```

```
text = """
artificial intelligence is transforming modern society.
it is used in healthcare finance education and transportation.
machine learning allows systems to improve automatically with experience.
data plays a critical role in training intelligent systems.
large datasets help models learn complex patterns.
deep learning uses multi layer neural networks.
neural networks are inspired by biological neurons.
each neuron processes input and produces an output.
training a neural network requires optimization techniques.
gradient descent minimizes the loss function.

natural language processing helps computers understand human language.
text generation is a key task in nlp.
language models predict the next word or character.
recurrent neural networks handle sequential data.
lstm and gru models address long term dependency problems.
however rnn based models are slow for long sequences.
"""
```

```
chars = sorted(list(set(text)))
char_to_idx = {c:i for i,c in enumerate(chars)}
idx_to_char = {i:c for i,c in enumerate(chars)}

vocab_size = len(chars)
print("Vocabulary size:", vocab_size)
```

```
Vocabulary size: 28
```

```
seq_length = 40
X = []
y = []

for i in range(len(text) - seq_length):
    X.append([char_to_idx[c] for c in text[i:i+seq_length]])
    y.append(char_to_idx[text[i+seq_length]])

X = np.array(X)
y = np.array(y)
```

```
model = Sequential([
    Embedding(vocab_size, 64, input_length=seq_length),
    LSTM(128),
    Dense(vocab_size, activation='softmax')
])

model.compile(loss='sparse_categorical_crossentropy',
              optimizer='adam')

model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument `input_lengt
  warnings.warn(
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	0 (unbuilt)
lstm (LSTM)	?	0 (unbuilt)
dense (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)

```
model.fit(X, y, epochs=20, batch_size=64)
```

```
Epoch 1/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 3s 65ms/step - loss: 3.3091
Epoch 2/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - loss: 2.9868
Epoch 3/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step - loss: 2.9699
Epoch 4/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 69ms/step - loss: 2.9335
Epoch 5/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 67ms/step - loss: 2.9359
Epoch 6/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 67ms/step - loss: 2.9136
Epoch 7/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 2s 122ms/step - loss: 2.9053
Epoch 8/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 94ms/step - loss: 2.9454
Epoch 9/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 2s 69ms/step - loss: 2.8527
Epoch 10/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - loss: 2.8378
Epoch 11/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 67ms/step - loss: 2.7253
Epoch 12/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 64ms/step - loss: 2.7064
Epoch 13/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 63ms/step - loss: 2.5992
Epoch 14/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 62ms/step - loss: 2.5494
Epoch 15/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step - loss: 2.5002
Epoch 16/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 66ms/step - loss: 2.4428
Epoch 17/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 2s 113ms/step - loss: 2.4068
Epoch 18/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 2s 117ms/step - loss: 2.4097
Epoch 19/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 65ms/step - loss: 2.3708
Epoch 20/20
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 71ms/step - loss: 2.3341
<keras.src.callbacks.history.History at 0x7c2da7fc6030>
```

```
def generate_text(seed, length=300):
    result = seed
    for _ in range(length):
        input_seq = np.array([[char_to_idx[c] for c in seed]])
        prediction = model.predict(input_seq, verbose=0)
        next_char = idx_to_char[np.argmax(prediction)]
        result += next_char
        seed = seed[1:] + next_char
    return result

print(generate_text("artificial intelligence is t"))
```

```
artificial intelligence is ters anet anet rone neters anet anet rone neters anet anet rone neters anet anet rone
```

```
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
```

```
tokenizer = Tokenizer()
tokenizer.fit_on_texts([text])
total_words = len(tokenizer.word_index) + 1
```

```
input_sequences = []

for line in text.split("\n"):
    token_list = tokenizer.texts_to_sequences([line])[0]
    for i in range(1, len(token_list)):
        input_sequences.append(token_list[:i+1])

max_len = max(len(seq) for seq in input_sequences)
input_sequences = pad_sequences(input_sequences, maxlen=max_len)

X = input_sequences[:, :-1]
y = input_sequences[:, -1]
```

```
from tensorflow.keras.layers import Layer, MultiHeadAttention, LayerNormalization, Dropout
```

```
class TransformerBlock(Layer):
    def __init__(self, embed_dim, num_heads, ff_dim):
        super().__init__()
        self.att = MultiHeadAttention(num_heads=num_heads, key_dim=embed_dim)
        self.ffn = Sequential([
            Dense(ff_dim, activation="relu"),
            Dense(embed_dim)
        ])
        self.norm1 = LayerNormalization()
        self.norm2 = LayerNormalization()

    def call(self, inputs):
        attn_output = self.att(inputs, inputs)
        out1 = self.norm1(inputs + attn_output)
        ffn_output = self.ffn(out1)
        return self.norm2(out1 + ffn_output)
```

```
embed_dim = 64

inputs = tf.keras.Input(shape=(max_len-1,))
x = Embedding(total_words, embed_dim)(inputs)
x = TransformerBlock(embed_dim, num_heads=2, ff_dim=128)(x)
x = tf.keras.layers.GlobalAveragePooling1D()(x)
outputs = Dense(total_words, activation="softmax")(x)

transformer_model = tf.keras.Model(inputs, outputs)
transformer_model.compile(
    loss="sparse_categorical_crossentropy",
    optimizer="adam"
)

transformer_model.summary()
```

Model: "functional_2"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 8)	0
embedding_1 (Embedding)	(None, 8, 64)	6,336
transformer_block (TransformerBlock)	(None, 8, 64)	50,048
global_average_pooling1d (GlobalAveragePooling1D)	(None, 64)	0
dense_3 (Dense)	(None, 99)	6,435

Total params: 62,819 (245.39 KB)
Trainable params: 62,819 (245.39 KB)
Non-trainable params: 0 (0.00 B)

```
transformer_model.fit(X, y, epochs=30, batch_size=32)
```

```
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 4.4211
Epoch 3/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 4.2475
Epoch 4/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 4.0977
Epoch 5/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 4.0003
Epoch 6/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 3.7707
Epoch 7/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 3.7079
Epoch 8/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 3.5582
Epoch 9/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 3.5093
Epoch 10/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 3.3965
Epoch 11/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 3.2792
Epoch 12/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 3.1976
```

```
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 2.9669
Epoch 15/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.8808
Epoch 16/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.7904
Epoch 17/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.6503
Epoch 18/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.5731
Epoch 19/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.4178
Epoch 20/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 2.3317
Epoch 21/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.1567
Epoch 22/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 2.1393
Epoch 23/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 2.0395
Epoch 24/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.9351
Epoch 25/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.8581
Epoch 26/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.7516
Epoch 27/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.6819
Epoch 28/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step - loss: 1.6574
Epoch 29/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.6288
Epoch 30/30
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 16ms/step - loss: 1.5508
<keras.src.callbacks.history.History at 0x7c2da7460050>
```